

WANGDONG JIA

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EDUCATION

University of California, Berkeley, CA

Ph.D. Candidate in Industrial Engineering and Operations Research

Sep 2022 – May 2027

(Advised by Prof. Xin Guo)

Research Interests: Stochastic control, Policy Learning, Optimization, Convergence analysis, Game equilibrium

Honor: Block Grant Award, IEOR Department Award (2023)

Relevant Coursework: Probability Theory, Statistical Learning, Optimization, Financial Engineering Systems

Columbia University in the City of New York, New York, NY

B.S. in Applied Mathematics and Applied Physics

Sep 2020 – May 2022

(Dean's list every year)

Honor: Pi Mu Epsilon, Math Honor Society (2021), Sigma Pi Sigma, Physics Honor Society (2021), Cum Laude (2022)

Relevant Coursework: Statistical Inference, Stochastic Process, Numerical Methods, Dynamical Systems

Centre College, Danville, KY

B.S. in Physics, Mathematics

Sep 2017 – May 2020

(Dean's list every year)

Honor: Winner of Putnam Mathematical Competition, national ranking 199 (2019)

WORK EXPERIENCE

Quantitative Strategist Intern, Morgan Stanley

May 2026 – Aug 2026

- Developed a model-free event-time framework for return volatility estimation in electronic FX, constructing meaningful clocks and selecting raw price series by price movements while suppressing microstructure noise.
- Designed generalized smooth filters across multiple frequencies to extract volatility signals with high spectral purity; implemented return and jump volatility estimators with near-instantaneous computation while reducing system load.
- Validated the framework on intraday FX data and macroeconomic events, showing event-time volatility tracks market evolution, adapts to changing activity, and recovers from price shocks with recovery times roughly two orders of magnitude shorter than wall-clock GARCH, reducing unnecessary defensive positioning and missed trading opportunities.

RESEARCH

Graphon Mean Field Game: Characterizing Nash Equilibria with Non-homogeneous Policy

May 2024 – Present

- Extending the Mean Field Game (MFG) framework by adapting the existing MF-OMO approach to non-homogeneous players and recasting the formulation in the Graphon MFG (GMFG) setting.
- Designed deep learning algorithms in Python/PyTorch using kernel methods, neural networks, and universal function approximation to compute GMFG Nash equilibria, including cases with multiple equilibria.
- Developed the Graphon Mean Field Games Occupation Measure Optimization (GG-OMO) framework that characterizes Nash equilibria of GMFG without the restrictive assumptions required in prior literature. Showed that suboptimal solutions still approximate equilibria stably under perturbations, providing theoretical robustness guarantees.

Medicare Overutilization Detection via Machine Learning Methods

Sep 2025 – Present

- Developed supervised and ensemble models (CatBoost, SVM, logistic regression) in Python to detect Medicare overutilization in ophthalmology; benchmarked classifiers using nested cross-validation and AUROC, with forward-backward stepwise feature selection.
- Developing an iterative semi-supervised learning framework for automated labeling.

Noise-Robust Estimation for Fractional Model

May 2021 – Mar 2022

- Developed a noise-robust converging estimator sequence for the Hurst index of fractional Brownian motion in the presence of normalized turbulence using pre-averaging techniques. Validated estimator using Monte Carlo in MATLAB.
- **Publication:** "Pre-averaging fractional processes contaminated by noise, with an application to turbulence," published in the *Journal of Applied Probability*. (<https://arxiv.org/abs/2212.00867>)

TEACHING EXPERIENCE

University of California, Berkeley

- INDENG 115: Industrial and Commercial Database Systems
- INDENG 172: Probability and Risk Analysis for Engineers
- INDENG 173: Introduction to Stochastic Processes
- INDENG 242B: Machine Learning and Data Analytics II

Sep 2023 – present

Instructor

Teaching Assistant

Teaching Assistant

Teaching Assistant

RELEVANT SKILLS

Programming & Software: Python (PyTorch, Polars, Pandas, CVXPY, Gurobi, scikit-learn), MATLAB